

Dengan data yang sudah diperbaiki:

$$x_0 = 78628734$$

$$x_1 = 1460962527$$

$$x_2 = 934458668$$

$$x_3 = 1985133557$$

$$x_4 = 721214858$$

$$\text{enc} = 93441898286175386159833534452283184887$$

sekarang data **konsisten sebagai LCG**.

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### 1. Hitung selisih

$$d_0 = x_1 - x_0 = 1382333793$$

$$d_1 = x_2 - x_1 = -526503859$$

$$d_2 = x_3 - x_2 = 1050674889$$

$$d_3 = x_4 - x_3 = -1263918699$$

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### 2. Recover modulus (m)

Hitung:

$$t_0 = d_1^2 - d_0 \cdot d_2$$

$$t_1 = d_2^2 - d_1 \cdot d_3$$

Hasil:

$$t_0 = 1175177090979332096$$

$$t_1 = 438459649889402880$$

Ambil GCD:

```
from math import gcd
```

$$m = \text{gcd}(t_0, t_1)$$

Hasil:

$$m = 2147483648$$

atau:

$$m = 2^{31}$$

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### 3. Recover multiplier (a)

Gunakan:

$$[a \equiv (x_2 - x_1)(x_1 - x_0)^{-1} \pmod m]$$

Python:

```
a = ((x2-x1) % m) * pow(x1-x0, -1, m) % m
```

Hasil:

```
a = 1103515245
```

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### 4. Recover increment (c)

Gunakan:

$$[c = (x_1 - a x_0) \bmod m]$$

Python:

```
c = (x1 - a*x0) % m
```

Hasil:

```
c = 12345
```

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### 5. Verifikasi

LCG yang ditemukan:

$$[X_{n+1} = (1103515245X_n + 12345) \bmod 2147483648]$$

Cek:

```
x0 = 78628734
```

↓

```
x1 = 1460962527 ✓
```

x1

↓

$$x_2 = 934458668 \checkmark$$

$$x_2$$

↓

$$x_3 = 1985133557 \checkmark$$

$$x_3$$

↓

$$x_4 = 721214858 \checkmark$$

Semuanya cocok.

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### Jawaban

$$a = 1103515245$$

$$c = 12345$$

$$m = 2147483648$$

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### 6. Prediksi output berikutnya

Hitung:

$$\begin{bmatrix} x_5 = (a \cdot x_4 + c) \bmod m \end{bmatrix}$$

Hasil:

$$x_5 = 1004965883$$

Jadi state berikutnya adalah:

$$1004965883$$

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Kalau challenge menggunakan pola:

$$\text{key} = \text{next\_lcg\_output}$$

$$\text{enc} = \text{bytes\_to\_long}(\text{flag}) \wedge \text{key}$$

maka langkah berikutnya adalah mencoba:

```
from Crypto.Util.number import long_to_bytes
```

```
enc = 93441898286175386159833534452283184887
```

```
key = 1004965883
```

```
flag = long_to_bytes(enc ^ key)
```

```
print(flag)
```

Kirim hasil outputnya jika belum berbentuk FLAG{...}, karena mungkin key yang dipakai adalah beberapa output LCG yang digabung, bukan hanya x5.